

Celebal Week 7 Assignment

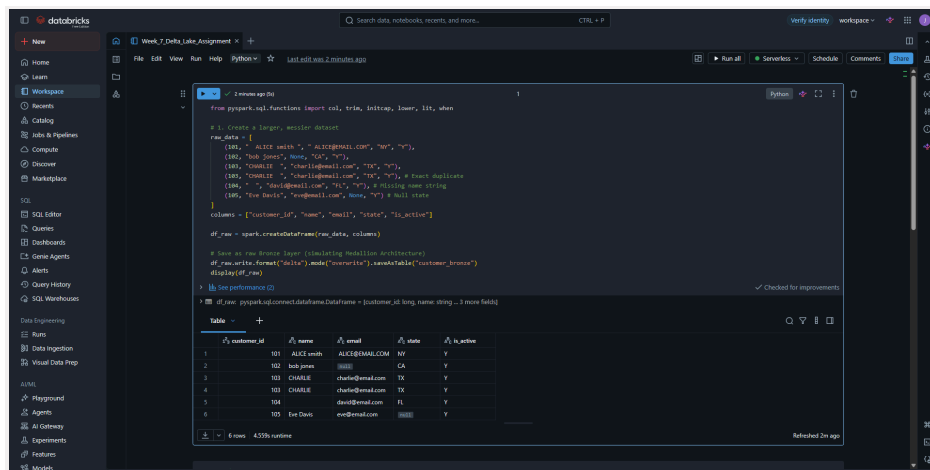
Objective:

To design and implement a robust data ingestion and cleaning pipeline using Apache Spark, culminating in a Delta Lake MERGE operation to handle incremental data updates.

1. Data Loading (Bronze Layer)

To kick off the pipeline, I put together a messy, raw dataset to mimic real-world incoming source records. I intentionally baked in all the classic data quality headaches, things like annoying leading and trailing whitespaces, all over the place capitalization, missing values, nulls, and duplicate entries.

From there, I saved everything straight into Databricks using the default Delta Lake format. Under the hood, a Delta table pairs standard Parquet files with a `\_delta\_log` directory full of JSON files. That transaction log is what gives you ACID compliance and handles concurrent writes safely, stuff you just don't get with plain Parquet or Hive tables. Sticking to a Medallion Architecture, this raw, completely untouched data forms our Bronze layer, making sure we keep the original state fully intact for data lineage and auditing down the road.



```
from pyspark.sql.functions import col, trim, ltrim, lower, ifnull, when

# 1. Create a larger, messier dataset
raw_data = [
    (101, "ALICE smith", "ALICE@GMAIL.COM", "NY", "Y"),
    (102, "Bob Jones", "None", "CA", "Y"),
    (103, "Charlie", "charlie@gmail.com", "TX", "Y"),
    (104, "Charlie", "charlie@gmail.com", "TX", "Y"), # Intentional duplicate
    (105, " ", "dave@gmail.com", "FL", "Y"), # Leading space + missing email
    (106, "Doe Davis", "dave@gmail.com", "None", "Y") # Null state
]

columns = ["customer_id", "name", "email", "state", "is_active"]
df_raw = spark.createDataFrame(raw_data, columns)

# Save as raw Bronze layer (preserving Medallion architecture)
df_raw.write.format("delta").mode("overwrite").saveAsTable("customer_bronze")
display(df_raw)

# Inspect performance
df_raw.spark.catalog.getTableMetadata(spark.catalog.getTableMetadata(df_raw.schema.catalogName).tableMetadata)
```

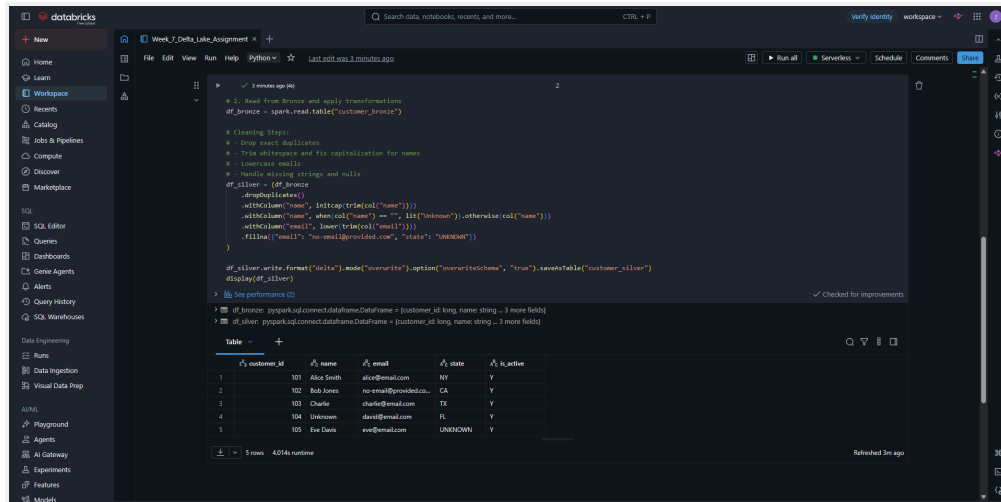
customer_id	name	email	state	is_active
101	ALICE smith	ALICE@GMAIL.COM	NY	Y
102	Bob Jones	None	CA	Y
103	CHARLIE	charlie@gmail.com	TX	Y
104	CHARLIE	charlie@gmail.com	TX	Y
105		dave@gmail.com	FL	Y
106	Doe Davis	dave@gmail.com	None	Y

2. Data Cleaning & Structuring (Silver Layer)

Following the multi-hop architecture pattern, the data was progressively refined from the Bronze layer into a cleaner, enterprise-wide Silver layer.

The following PySpark transformations were applied to structure the data:

- Deduplication: Dropped identical rows to ensure data integrity.
- String Manipulation: Applied `trim()` and `ltrimcap()` to standardize names, and `lower()` to standardize email addresses.
- Null Handling: Replaced empty strings with a default "Unknown" string and used `.fillna()` to provide default values for missing emails and states.

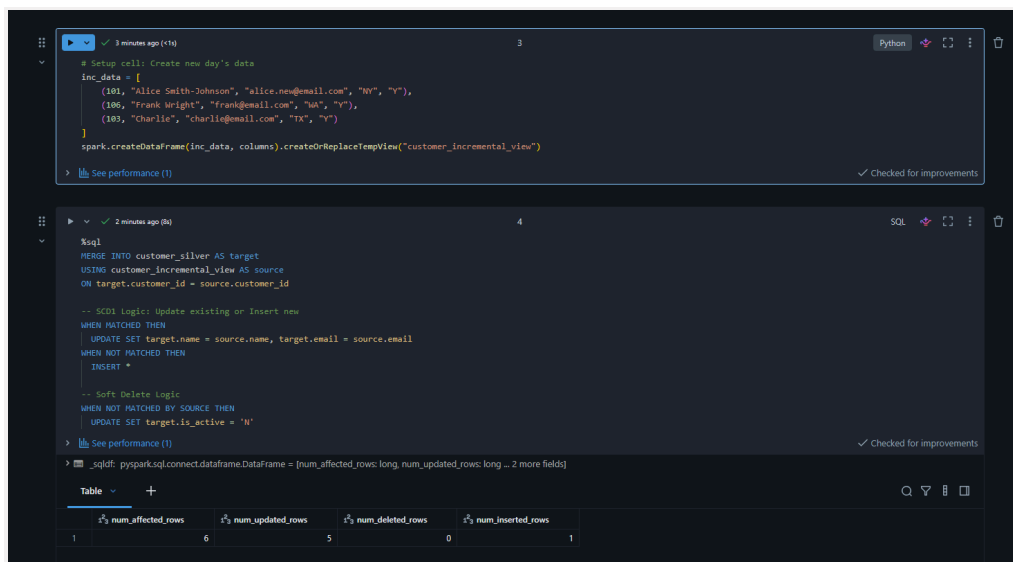


3. Incremental Upserts (SCD1) & The MERGE Operation

To test the pipeline's ability to handle evolving data over time, a second dataset (`customer_incremental`) was introduced representing a new day's data.

A SQL `MERGE` statement was used to upsert this incremental data into the target Silver table. The operation relied on up to three distinct matching clauses to determine the outcome of each row:

- **WHEN MATCHED:** Implemented Slowly Changing Dimension Type 1 (SCD1) logic by updating the target row's email and name with the new values from the source, intentionally overwriting the old state with no history kept.
- **WHEN NOT MATCHED:** Inserted entirely new customer records that only existed in the source data.



4. Historical Preservation via Soft Deletes

Instead of executing a hard `DELETE` for records that dropped out of the source system, the `MERGE` operation utilized the third matching clause.

- **WHEN NOT MATCHED BY SOURCE:** Handled keys that were present in the target but absent from the source.

This clause was used to implement a soft delete by updating an `is_active` flag to 'N' instead of physically erasing the row. This approach preserves historical data and ensures past records remain fully queryable for auditing purposes.



```
# Validate the Soft Delete (Records missing from source are flagged, not deleted)
df_soft_deleted = spark.read.table("customer_silver").filter(col("is_active") == 'N')

# Take a screenshot of this output showing the 'N' records
display(df_soft_deleted)
```

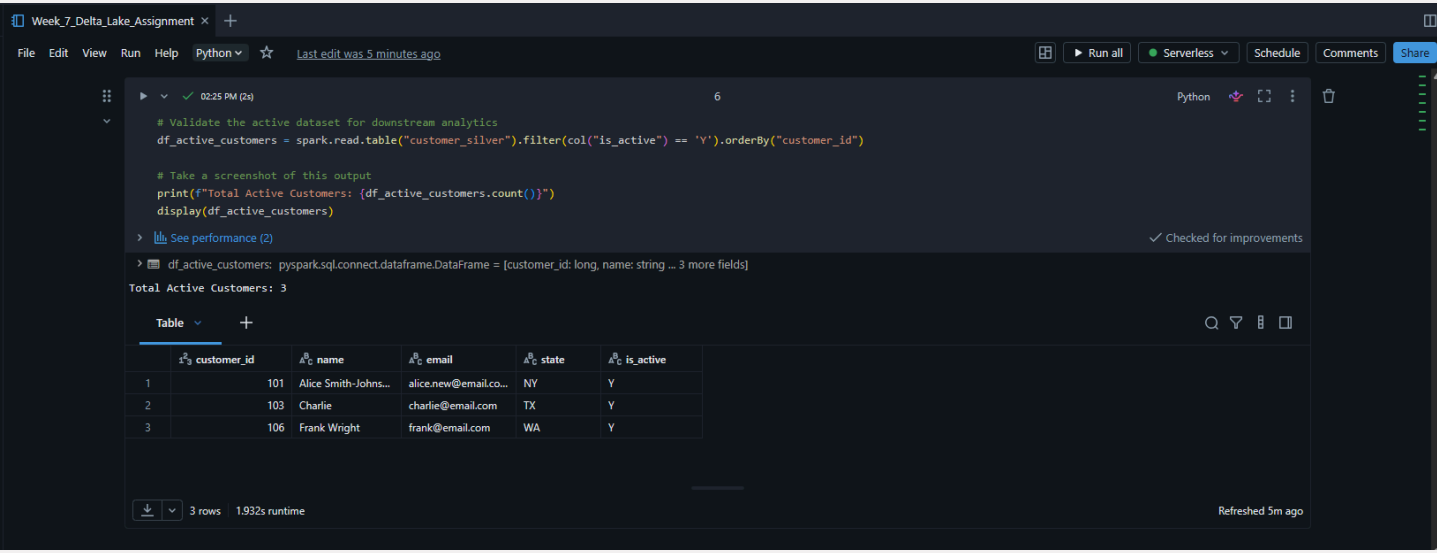
Table

	customer_id	name	email	state	is_active
1	104	Unknown	david@email.com	FL	N
2	102	Bob Jones	no-email@provided.co...	CA	N
3	105	Eve Davis	eve@email.com	UNKNOWN	N

3 rows 1.363s runtime

5. Validation and Final Output

To confirm the success of the pipeline, the dataset was queried to validate the downstream impact. The current, active state of the table can easily be accessed by analysts simply by filtering for `is_active = 'Y'`.



```
# Validate the active dataset for downstream analytics
df_active_customers = spark.read.table("customer_silver").filter(col("is_active") == 'Y').orderBy("customer_id")

# Take a screenshot of this output
print(f"Total Active Customers: {df_active_customers.count()}")
display(df_active_customers)
```

Total Active Customers: 3

Table

	customer_id	name	email	state	is_active
1	101	Alice Smith-Johns...	alice.new@email.co...	NY	Y
2	103	Charlie	charlie@email.com	TX	Y
3	106	Frank Wright	frank@email.com	WA	Y

3 rows 1.932s runtime

The final dataset accurately reflects the updated information for existing keys, the insertion of the new keys, and the successful application of the 'N' flag for stale records.

Week_7_Delta_Lake_Assignment

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Display the entire final dataset (both active and inactive records)
df_final = spark.read.table("customer_silver").orderBy("customer_id")

Take a screenshot of this output
display(df_final)

See performance (1)✓ Checked for improvements

> df_final: pyspark.sql.connect.dataframe.DataFrame = [customer_id: long, name: string ... 3 more fields]

Table

	customer_id	name	email	state	is_active
1	101	Alice Smith-Johns...	alice.new@email.com	NY	Y
2	102	Bob Jones	no-email@provided.co...	CA	N
3	103	Charlie	charlie@email.com	TX	Y
4	104	Unknown	david@email.com	FL	N
5	105	Eve Davis	eve@email.com	UNKNOWN	N
6	106	Frank Wright	frank@email.com	WA	Y

6 rows1.303s runtime

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